Pre-Feasibility Study of Alfalfa Forage Crop with Pivot Irrigation System in Ma'an Governorate

2017

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1. Executive Summary

The project aims to cultivate and produce green forage (Alfalfa), partially drying and pressing in bales weighing about 17-18 kg and selling it to livestock breeders, especially dairy cows herders throughout the Kingdom, mainly to dairy cattle breeders in Al Halabat area in Zarqa Governorates to cover the increasing annual demand for forage crops (clover and Alfalfa). Where the imported amount of clover was 40,000 tons of worth about 8.5 million dinars in 2015.

The following table shows the main results of the study:

Table 1 Main result of the study

Project Name	Alfaalfa Forage Crop
The Proposed Site	Ma'an
The Products of the Project	Dried Alfalfa Bales
Total Labor	9
Total Investment (,000 JD)	715.6
Net Present Value of Cash Flows (,000 JD)	2,208
Payback Period (Years)	3
Internal Rate of Return	%38.5

The investment cost of this project was estimated at 715,000 JD. This amount is determined through fixed assets which include: The cost of the pivot irrigation system, the drilling of groundwater wells, the initial land prepration for alfalfa cultivation, irrigation systems, machinery and equipment, furniture and transport, pre-operating expenses and initial working capital. (40%) of the project will be financed through loans and the rest through self financing. In the case of the availability of treated water from the purification plants, the project becomes more useful, and despite the high cost of investment for the project does not provide many employment opportunities commensurate with the size of investment because of the adoption of the project on agricultural mechanization in agriculture and irrigation and crop collection and sale. The results of the financial analysis indicate that the net present value of the cash flows of the project is positive, which is indicative of the financial feasibility of the project and the internal rate of return is much higher than the weighted average cost of capital, which is indicative of the feasibility of the project. Other financial indicators indicate that the project is profitable to the investor.

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

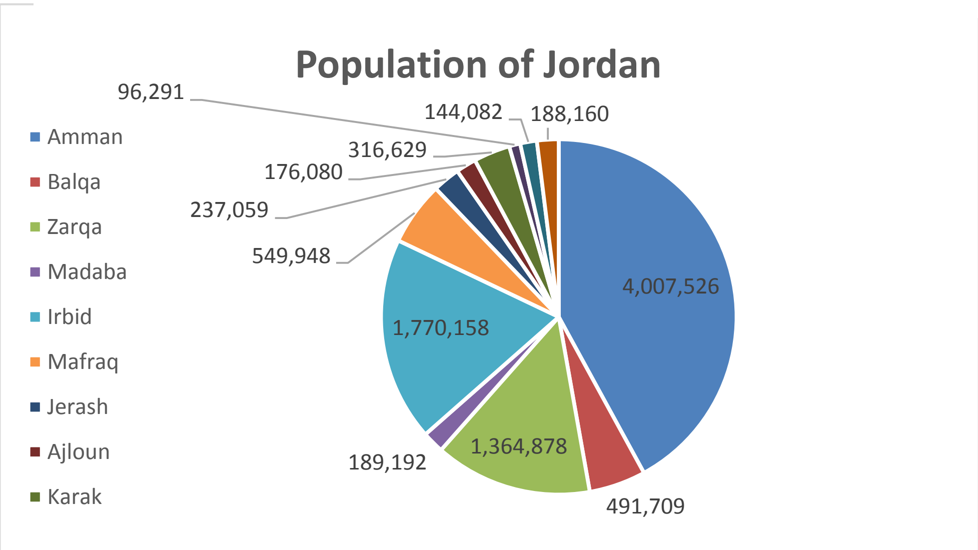
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

Department of Statistics (DOS), 2015.

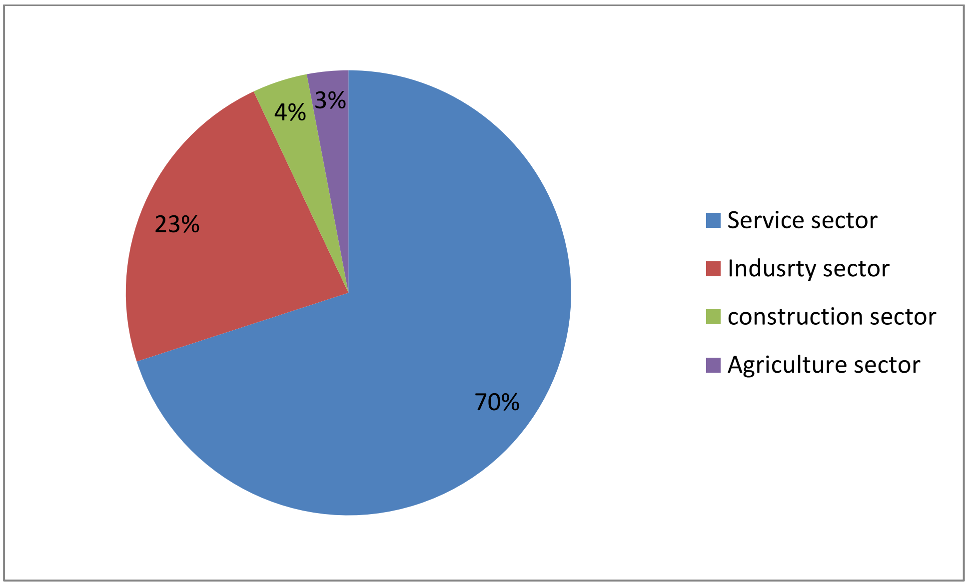
Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law).
These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is

2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The project is based on the idea of cultivating 1000 sqm area of clover (Alfalfa) in an axial irrigation method, drying and pressing with special machines and equipment and selling it to the milk cows farmers in Jordan. As these projects are considered complementary agricultural projects linked directly to the production of milk and dairy products, and these projects can be established provided the availability of surface or groundwater and preferably cultivation on the treated water from the purification plants, which often require high capital, the project is one of the commercial related projects that do not require a large labor, where the machines and equipment are heavily relied on shearing, drying, pressing and handling.

Project Objectives

The Jordanian market needs this material in huge amounts, as the annual average value of imports during the period (2010-2015) is about 6 million dinars. The project aims to:

1. Achieving a profitable return to the owner of the project,
2. Cultivating and producing green clover (Alfalfa) and supplying it to cattle breeders with distinctive quality, good and mold free,
3. Providing jobs in the regions, and exploiting local sources of water and unused land
4. Cover the annual deficit of alfalfa, which is imported from unreliable neighboring markets.

Proposed Product

The demand for feed, especially alfalfa is derived from the demand for fresh milk and dairy products. Green feeds are as important as other food. They are very important for the animal, especially the pregnant animal, because it contains vitamins and mineral salts. Green clover (Alfalfa) is rich in vitamin A, which is important for cows. As there are no available quantities of alfalfa, the quantity provided to dairy cows is about 10-12 kilo green clover / day, and the pregnant cows about 15 kg of green clover / day, for its importance for the fetus and especially the last 30 days of pregnancy. The feed needs of the cattle sector represent more than 72% of the cost of production. Therefore, the recent increase in feed prices has led to an increase in the cost of milk production. The cost of milk production per liter is from 360-400 fils in commercial farms.

The need for fodder is very necessary for cows because it works to fill the stomach to feel satiation. It also helps the cows in the process of digestion and activation of the cells of the digestive system in addition to the balance of the composition of fatty acids by microorganisms in the stomach and thus fodder fills a significant contribution to the composition of fat in milk resulting from the proportion of concentrated feed and fodder in order to maintain the natural activity of the stomach as well as the percentage of natural fat in milk. If the percentage of concentrated feed exceeds the required limit (60%) and the percentage of filler is less than 40%, this reduces the fat content in milk produced significantly, the requirement of the dairy cow from the filler depends on its production and size, but it can be said that the filler provides an average of 8-10 kilo / day from dry feed.

Market Size Analysis

The data of the department of Statistics indicate that the average area cultivated with Alfalfa is about 60 thousand dunums in 2015 with a quantity about 204 thousand tons (Table 2). The production of these areas of clover (Alfalfa) is unable to cover the needs of livestock breeders, allowing traders to import this material from Spain and Egypt in the recent period after it was imported from Saudi Arabia, which stopped the export of fodder.

Table 3 : The development of cultivated area and production of alfalfa in Jordan

Year	Cultivated Area (Dunums)	Production (Tons)	Production (Tons/Dunum)	Total irrigated area in Jordan (Dunums)	Percentage of cultivated area by alfalfa to total irrigated area
1990	8,275	8,932	1.08	630,030	1.31%
2000	15,894	59,566	3.75	769,116	2.07%
2010	65,778	223,591	3.40	1,024,721	6.42%
2011	22,100	100,323	4.54	964,495	2.29%
2012	36,604	148,259	4.05	956,538	3.83%
2013	49,564	232,409	4.69	1,034,433	4.79%
2014	66,833	277,169	4.15	1,050,471	6.36%
2015	60,110	244,223	4.06	1,034,801	5.81%
AVG	**50,165**	**204,329**	**4.15**	**1,010,910**	**4.92%**

Table 3 shows the value and quantity of imports of alfalfa during the period (2011-2015). The average quantity of imports of dried clover was 22,000 tons with an estimated value of 5.4 million dinars. About 40,000 tons of clover (Alfalfa) was imported in 2015 with a value of 8.5 million dinars, the table shows that there is a deficit in the trade balance of the decimals for the year 2015 reached about (8.4) million dinars.

Since the demand for alfalfa is a demand derived from the demand for fresh milk produced by milk cows in specialized farms, it is necessary to address the demand for milk and the preparation of cows and distribution in the Kingdom.

The total production of the kingdom of milk amounted to about 462 thousand tons, the contribution of cows sector is about 352 thousand tons by the amount (76%) of the local production of the Kingdom of milk, which is considered a major source for the manufacture of food products, and there are broad fields to Invest in this sector as the local production of milk does not cover more than (78%) of the size of the needs of local consumption, knowing that the demand for milk and dairy products continues to increase as a result of the increase in population and prices compared with other products and the increasing in nutritional awareness in terms of the importance of milk in human nutrition and multiple Forms of products that fall in most diets.

Table 4 : Value and quantity of imports, exports, re-exports and trade deficit of alfalfa

Year	The value of Imports (000 JD)	Imports Quantity (Tons)	The Value of Local Exports (000 JD)	Local Export Quantity (Tons)
2011	3,389	14,006	242	-2,893
2012	3,302	12,503	264	-3,271
2013	5,499	19,492	282	-5,473
2014	6,403	24,923	257	-6,378
2015	8,516	39,725	214	-8,349
AVG	**5,422**	**22,130**	**252**	**-5,273**

Source: Department of Statistics (2017) Foreign Trade Reports

The number of dairy cows in 2015 was about 66,880 heads, 60,315 heads of them were a Dutch dairy cow and 6,565 were domestic dairy cow as shown in the following table. Cows breeding are widespread in most of the Kingdom's governorates and are concentrated in four governorates: Zarqa, Mafraq, Irbid and the capital (Table 4). There are (18,915) family breeding, mostly in Irbid governorate.

Table 5 : The Number of Livestock for 2015

Breed	Cows/Mothers	Veal	Calves	Bulls	Total
Domestic	6,565	1,500	1,143	150	9,358
Dutch	60,315	6,437	4,997	568	72,317
Total	66,880	7,937	6,140	718	**81,675**

Source: Ministry of Agriculture (2017). Annual Report of the Directorate of Animal Resources, 2015.
Ministry of Agriculture, Amman, Jordan.

The most important problems facing cow raising and milk production are the high and fluctuating prices of fodder, which are mostly imported from outside jordan. The increase in fodder prices is one of the most important obstacles facing this sector through the direct impact on the farms and the product. As the cost of fodder are about 72% of the production inputs.

The proposed scientific means to overcome these constraints is to increase the local production of fodder and alfalfa forage crops and encourage their cultivation to reduce the pressure of high prices on cows' breeder.

Jordan relies on providing the bulk of its fodder and alfalfa forage crops needs on imports. The private sector imports soybeans, maize, feed concentrates of animal and plant origin, animal feed, various plant gravy and feed additives, which are incorporated into the feed mix, as well as straw, hay and alfalfa. The Ministry of Industry and trade has imported barley material and ensure that it is properly distributed to livestock breeders at the subsidized price. In the year 2015, about 1,970,106 tons of feed were imported; 44,675 tons of it was alfalfa.

The amount of domestic demand on the ready-made alfalfa was determined based on the review of the quantity of exports and imports and the quantities of re-exports in addition to the local production estimated by the Department of Statistics. Thus, the size of the market was estimated on alfalfa during the period 2011-2016 as shown in Table 5.

Table 6 : Market Size of Alfalfa during (2011-2016)

The Market Size of Alfalfa	2011	2012	2013	2014	2015	2016
Imports (Tons) (1)	14,006	12,503	19,492	24,923	39,725	40,520
Exports (Tons) (1)	62	12	0	0	0	0
Re-exports (Tons)	953	93	95	220	2,927	2,986
Local Production (Tons) (2)	100,323	148,259	232,409	277,169	244,223	249,108
Consumption (Tons) (2)	113,314	160,657	251,805	301,871	281,021	286,642

(1) Source: Department of Statistics (2017) Foreign Trade Report.

(2) Source: Department of Statistics (2017) Agricultural Report.

Competitors Analysis

Local production of alfalfa is estimated at about 244 thousand tons in 2015 and alfalfa is produced in a number of locations in Jordan, including Mafraq, Ma'an and Zarqa (Table 6), as well as around most of the purification plants in other governorates. The market has a number of specialized producers in the supply of alfalfa to dairy farms.

Therefore, it is not expected that the project will face difficulties in achieving the expected sales because the alfalfa is imported from Spain and Egypt and the buyer from the cow breeders prefer to buy the domestic product periodically and continuously over the imported alfalfa, because the quantity imported must be large and requires a warehouse or Storage places in addition to administrative transactions related to import.

Prices Analysis and Pricing Policy

Clover (Alfalfa) is sold through direct contracting between the producer and the livestock breeders at a minimum price (280) JD/ton on the farm land during the summer period, where the production increases and reaches a higher level to (330) JD/ ton during the winter at the farm gate price, Buyer affords transportation costs to the site of consumption or storage.

The project aims to produce a high-quality protein free of mold and disease in accordance with Jordanian approved standards.

Table 7 : Area and production of Al-Falfa in 2015 by place of producion

Region	Cultivated Area (Dunums)	Percentage	Production (Tons)	Percentage	Productivity (Tons/Dunum)
Az Zarqa	9,255	15%	36,905	15%	3.99
Irbid	50	0%	165	0%	3.30
Al-Mafraq	26,460	44%	106,955	44%	4.04
Ma'an	14,485	24%	58,142	24%	4.01
Al-Aqaba	4,717	8%	17,926	7%	3.80
Al-Shuna Al-Shmalyiah	1,145	2%	6,343	3%	5.54
Deir-A'llaa	1,821	3%	8,276	3%	4.54
Al-Shuna Al-Janubyiah	1,828	3%	8,261	3%	4.52
Ghoor Al-Safi	347	1%	1,251	1%	3.60
Total	**60,110**	**100%**	**244,223**	**100%**	**4.06**

Market Share

Local demand for ready-made alfalfa was estimated based on the study of the needs of dairy cows of green material, and average consumption of cow milk was estimated by about 12 kg per day, assuming that 82% of the herd are dairy cows and 18% of the rest of the herd consumption rate 6 kg per day Of alfalfa, assuming a growth rate of 2.0% per year in the number of dairy cows for the coming period. For the period of 2017-2026, the demand of alfalfa was predicted by multiplying the daily requirements of the milking cows by the number of days of the whole year (365) and then multiplying the expected number of cows (assuming constant consumption of one cow of clover). The future demand on alfalfa during the period (2017-2026) is shown in (Table 7).

The project's market share is about 2500 tons per year and it can be determined from the cultivation of 1000 dunums in the desert lands in Ma'an Governorate, based on the expected productivity of the project which is 2.5 tons / year of dried alfalfa by solar drying by 8-10 seedlings per year of green clover (Alfalfa), with an average yield of 1.5-2.0 tons / dunum of green clover. The ratio of dry alfalfa to green is 1: 5, according to technicians in this field.

Thus, the project's share can be estimated based on the expected production capacity of the project, which amounts to less than 1% of the total demand for alfalfa.

Estimated Revenues Analysis

The clover (Alfalfa) is harvested at an average of 8-10 times per year. The dry clover production rate in open desert lands is 2.5 tons / dunum. Alfalfa is a perennial plant, but it is preferred to be renewed every five years, after which productivity begins to decline. The selling prices of dried alfalfa range from (280-330) JD/Ton by the agricultural season, and sold in the form of bales weighing about 18 kg.

Table 8 shows the projected revenues from the project. For the purposes of this study, it was considered that the average sale price 280 dinars / ton on the farm door after the process of knocking and equipment in bales is expected to increase prices by 2% per year.

Table 8 : The domestic market share during the period (2018-2027)

Quantitative Demand Indicators for Alfalfa	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Number of expected dairy cows (Head)	58,060	59,337	60,643	61,977	63,340	64,734	66,158	67,614	69,101	70,621
Annual head consumption rate (kg / head)	4,380	4,380	4,380	4,380	4,380	4,380	4,380	4,380	4,380	4,380
The apparent demand for milk on the alfalfa in tons	254,303	259,898	265,616	271,459	277,431	283,535	289,773	296,148	302,663	309,321
Number of expected non-milking cows (Head)	12,745	12,746	12,747	12,748	12,749	12,750	12,751	12,752	12,753	12,754
Annual head consumption rate (kg / head)	2,190	2,190	2,190	2,190	2,190	2,190	2,190	2,190	2,190	2,190
The apparent demand for the rest of the herd on the alfalfa in tons	27,911	27,914	27,916	27,918	27,920	27,923	27,925	27,927	27,929	27,931
Total expected demand on the alfalfa	282,215	287,811	293,531	299,377	305,352	311,457	317,697	324,075	330,592	337,253
Expected Project Production (Ton)	1,250	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500
Project market share of domestic demand (%)	0.4%	0.9%	0.9%	0.8%	0.8%	0.8%	0.8%	0.8%	0.8%	0.7%

Table 9 Estimated revenues during (2018-2027)

Projects Revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Sales (Tons)	1,250	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500
Price (JD/Ton)	280	286	291	297	303	309	315	322	328	335
Revenues (JD)	**350,000**	**714,000**	**728,280**	**742,846**	**757,703**	**772,857**	**788,314**	**804,080**	**820,162**	**836,565**

Alfalfa and its scientific name spp Medicago Sativa, a plant that is used as feed and belong to the legume family. Has a life span of 5-12 years depending on the used quality It reaches a height of 1 m and develops dense masses of small purple flowers. The roots are usually very deep and can reach 4.5 meters. This enables the plant to be particularly resistant to drought.
Like all legumes, roots have nodules containing rhizomebium bacteria, with the ability to stabilize atmospheric nitrogen, and produce high-protein food, regardless of the nitrogen available in the soil. Its ability to stabilize nitrogen (increase nitrogen in soil) and its use as animal feed improves the efficiency of agriculture. Alfalfa is the queen of legumes and requires high temperatures and dry weather in summer. Spain and Argentina are two of the best countries in the world to produce clover.

Vegetation description

Clover is a perennial plant and is harvested between 8 and 10 times a year depending on the climate, soiland careness. Clover is a herbaceous plant with a mullet remains in the soil for a period of up to fifteen years. It is recommended that it be kept for no more than five years. After which productivity begins to decline. The flowers are found in cluster shoots, the flower is fluffy and the color of the flowers ranges from green to purple by type and variety. In vitro fertilization is mainly dependent on insects. The seeds are found in helical horns. The number of horn rolls varies from 2-3 laps by type, variety and seed of small kidney size, from olive green to light yellow depending on the duration of storage after harvest.

Food characteristics

Alfalfa is an animal feed plant with distinction. As a source of animal feed, it has excellent feeding properties, including:

- High protein content. Alfalfa, unlike meat meals, provides a large amount of vegetable protein, which will improve the health of animals and people. Alfalfa is an excellent green alternative to animal feed. High in other nutrients such as nitrogen, phosphorus, potassium, calcium, boron, sulfur, magnesium, molybdenum
- Fiber-rich ratio: Fiber intake in animal feed depends on the size of the particles in the given crop. The fiber components are fermented by the gastrointestinal microorganisms of ruminants. This causes the digestion to be generally good.
- A high fiber digestion factor and a large amount of amino acids and milking cells will eat more dry material and will produce more milk when fed with high NDF.
- The addition of large-scale foods in the ruminant diet is necessary to stimulate the digestive system and to maintain their health and productivity from milk.

- The process of drying clover reduced the loss of nutritional value (leaves, protein and vitamins) and the risk of pollution from the soil, while avoiding moisture from dew or rain that would increase microbial contamination. Drying, as a means of maintaining fodder, causes almost no changes in its nutritional value. There is little, but to a minimum, decrease in organic material digestion (OM) but no change in energy value. There is a change in digestive capacity and food absorption in the animal.
- Alfalfa contains about 50% in its composition of the cell wall and a balanced fiber composition (8% pectin, 10% hemicellulose, 25% cellulose and 7% lignin) and therefore ensures rapid digestive passage, large amount of soluble fiber and tampon) High, this, along with a high palatability of taste, makes alfalfa an option to be a major component in the feed of milk cows

- The content of the crude protein (CP) is largely determined by its market value. The earlier it was collected, the lower the production of dry matter (DM) per dunum. On the other hand,
- It is estimated that (CP) is a good indicator of the value of food energy, so that a single increase of 1% of the CP on dry material represents a 0.03 increase of UFL (net energy secretion of milk).

Central pivot Irrigation

Pivot sprinklers are one of the most important modern irrigation systems where possible through cultivation of vast tracts of land and limited quantities of water, especially desert lands and central spray is the irrigation machine and it has a special mechanism. There are many companies that produce these central sprinklers, and central sprinklers are made up of several towers. A number of towers are not required for central sprinklers, but the number of towers depends on the space available to us, which the central machine can be built on. The length of towers varies according to the company produced, towers up to 54 meters long like Vali and Falmont and towers up to 52 meters long, such as Zimatic and towers 48 meters like Aakl.

Each tower consists of several pivot towers, which are pipes usually 6 inches in diameter and vary in diameter in some types, such as the familmont, with a certain number of towers up to 8 inches in diameter. The number of these towers is usually the first of the five towers of the sprinkler. This is in order to push a larger quantity of water to the external towers. The tower is also composed of a number of sprays, which are often 28 sprays or small submersibles, and different diameters of sprayers or sprinklers from one tower to another. In the internal towers there are small openings. Then these openings gradually increase until we reach the last tower of the central machine. , Each tower consists of 2 gearboxes mounted in the 2 frames for the sprinkler movement.

Each tower consists of 1 center motor, which is an electrically powered motor and controlling the movement of the gears inside the gearbox through the movers and then rotate the frame and move the central machine, each tower consists of a pair of links called Kobling, From center motor to gearbox, each tower in the sprayer consists of 7 angles to maintain the sprocket balance. The first angle in the tower, like the last angle, extends the lengths of the angles until the maximum length reaches the angle in the middle of the machine.

Each tower is connected to the next tower by a joint or a piece of plastic. The central machine works with the electric power by 480 volts and derived either from an electrical source if the electricity is in the region or from a generator to generate electricity from the main machine. Each tower has two tires are usually 24 * 14.9 sizes and air pressure in tires is 35 bar.

The machine base consists of four-legged, mounted to the ground at the beginning of the machine and the base in the middle of the area to be cultivated mostly, and sometimes be in the side if the machine is semicircle only, there is a control panel in the machine at the base; a plate through which control the movement of machine and speed, there is in the control panel a key used to move the sprayer forward and another to move it back, there is also key to connect the power supply, and there are also remote control at the speed of sprayer, and there are ammeter to measure the strength of the electric current, there is at the top of the base a circular piece called The cricket ring transporting the Electric current to the rest of the parts of the sprayer.

The speed of the sprinkler is determined by the percentage, ie, the speed control device is divided into 10% 20% 30% 40% 50% to 100% and the speed of 100% means that the sprayer does not stop and the whole cycle is running and the last tower stops spinning. The last tower is connected to the one before it and there is a small arm called the balance column at the last movement of the tower to a certain extent, the balalnce column moving with it and then to a certain extent of the movement presses the balance column on the key in the previous tower and moves the previous tower and so each tower is between it and the tower followed by the balance column and it is an iron skewer about 60 cm long and about 6 mm thick.

Raw Materials Requirements

There are many agricultural processes required to produce alfalfa starting from soil preparation, tillage, leveling, agriculture, fertilization, pesticide spraying, and irrigation, ending with crop cutting, drying and pressing in bales form. The required raw materials for the production of clover (Alfalfa) are seed, organic manure, chemical manure of different types, pesticides and resistance to weeds, and irrigation water.

It is known that alfalfa consumes large amounts of water compared to other agricultural crops due to its permanence in the soil and the density of vegetation and the speed of development, the average water needs is about 1440 cubic meters per year for one dumun. Table 9 shows the cost of raw materials for agriculture and the annual cost of planting 1000 dunums of clover under the central irrigation system

Table 10: Prices, quantities and cost of annual raw materials required

Raw Material	Unit	Price/ unit	Quantity/ Dunum	Value JD/ Dunum	Annual Quantity (Unit)	Annual Cost (JD)
Seed planting	KG/ Dunum	9	5	45	5,000	45,000
Organic fertilizers	Ton/ Dunum	50	1	50.0	1,000	50,000
chemical fertilizers	KG/ Dunum	0.5	20	10.0	20,000	10,000
Pesticides	L/ Dunum	15	1	15.0	1,000	15,000
Others	JD/ Dunum	20	1	20.0	1,000	20,000
Electricity for pumping water	m^3/ Dunum	0.033	1440	48.0	1,440,000	48,000
Cables and Wires	Roll/ Dunum	0.12	138	16.6	138,000	16,560
Total				**204.6**		**204,560**

4. Technical Analysis

The Required Human Resources

This project requires a plant production agricultural engineer specialized in the production of fodder in addition to an agricultural technician as well as an employee to carry out the product marketing, accountant, agricultural workers and drivers of agricultural tractors. The following table (Table 9) shows the employment requirements.

Table 11 : The Required Labor

Job Title	No.
Project Manager -Agricultural Engineer/Food Processing	1
Agricultural Irrigation Technician	1
Marketing representative and accountant	1
Guard	1
Total Indirect Employees	**4**
Maintenance Technician	1
Cultivation workers	2
Tractors Driver	2
Total Direct Employees	**5**
Total Employees	**9**

Job Description

The agricultural Engineer: should have sufficient knowledge of type of forage crops mainly alfafa, planting date, technical ability of secure raw materials and ordering primary inputs for plant cultivations, land preparation, levelling, primary tillagem adding organic and chemical fertilizers, installing irrigation system, irrigation schedualing, procedures of plant protections and pest control. Forage cutting, drying, assembling and baling, transport.

Agricultural Engineering: Complete knowledge of alfalfa varieties and their date of planting, the ability to request and provide the necessary materials of production tools and requirements, the ability to prepare the agricultural land for growing the crop, such tillage preparation, leveling, fertilizing application, adding organic fertilizer, implementation of prevention and pest control measures in accordance with environmental standards. Harvesting the product and assembling and drying products. Processing and compressing the products, supervising the cutting, drying, pressing and transportation process, filling the farm records and work models, documenting the expenses and preventive control procedures and schedules, proposing production policies and action plans, monitoring their implementation and evaluating their results. Organizing the daily work, distributing the employees to different tasks, following up the implementation of their work, coordinating with the different production parties to accomplish the required tasks as planned. Determine the types of alfalfa and clover suitable for planting, identify the types of pesticides needed for different types of plants, their sparing time and quantities, in addition to determining and securing the quantities of fertilizers required in coordination with technical specialists. Coordinate with the transport companies to ensure the sales and supplying production and achieve the appropriate return, verify the accuracy of quantities delivered. Check the availability of requirements and requirements for the safety and security of stored materials from pesticides, organic and chemical fertilizers to not harm the environment and public health, making proposals aimed at improving the level and quality of production. Supervising the technical and management of the staff bad labors, evaluating their performance and achievements and proposing programs to raise their knowledge and develop their abilities. Develop procedures and ensure safety and occupational healt

Agricultural Technician: land preparation for the cultivation of the crop by tillage, laying and softening. Provide materials and work equipment of seeds, seedlings, fertilizers and pesticides according to the quantities and specifications specified. Planting the seeds according to species and varieties, preparing and setting the specifications of the raw materials and the appropriate time schedule for their supply, following up the procedures of maintenance of the machines and preparing the appropriate factors and drivers for them, preparing programs for maintenance of machines and equipment, identifying materials for the production plan, Preparing the necessary equipment for the production plan and providing it in the warehouses with their quantities and schedule for their use, and developing appropriate technical solutions for the problems related to the production process, knowledge of how to store the product and raw materials, and developing the procedures and methods of work and production using modern technologies. Preparing technical reports on agricultural and production operations, and implementing procedures and securing occupational and public health and safety requirements.

Marketing Officer: Preparation for receipt of product and matching quantities, types and specifications with the purchase order, and check its suitability for animal consumption, monitor the packaging process of products and delivery system, and follow-up loading and unloading of the products, bargaining on selling prices, clarify the advantages and specifications of the different product types, and help customers to choose their needs , and to respond to their queries, and securing their applications and follow-up processing, and the weight of materials, product sale, customer accounting, billing, monitor the inventory, determine the needs of the establishment of agricultural production requirements, and preparing lists of shortage of raw materials. Organization of sales records and financial records. following the procedures and instructions of occupational health and safety.

Agricultural worker: works alone or under the supervision of the agricultural engineer and prepares the necessary materials, preparing the land for cultivating by plowing, leveling, planting the seeds and caring for the crop, install irrigation system, and spraying the plants with pest control materials such as insects, diseases and rodents in accordance with environmental standards, protecting the crop from the risk of frost. participation in picking and harvesting products, packing, packaging, preserved and stored. Application of occupational safety and health procedures and instructions.

Location Analysis

Alfalfa plant is a plant that bears various environmental conditions. Where it bears low temperatures to the extreme frost, high temperatures in the summer. And the higher the quantity of water and fertilizer given to alfalfa, the better quality and higher quantity of alfalfa produced. Therefore, it is prefered to grow in Ma'an governorate, beside water treatment plants. The Ministry of Water reports indicate that the quantity produced by the Ma'an treatment plant is partially exploited and there are plans to increase the number of people employed by the sewage system. It is expected that 1.3 million cubic meters of reclaimed water will be discharged after the implementation of the sewage network. And currently exploited from the reclaimed water at the Ma'an plant does not exceed 250 thousand cubic meters of nearby farms, which is expected to be a surplus estimated about one million cubic meters is not exploited.

In any case, the financial analysis of the project was based on the assumptions of allowing the drilling of two wells to access underground water in the desert areas within the Ma'an Governorate in areas that are not water-depleted, and this project does not constitute an additional burden on the reservoir of groundwater. Most of these areas are remote areas with few populations rich in unexploited groundwater resources. Based on all these data, it is proposed to establish a number of projects for the production of clover in the Ma'an area and in agricultural areas where there is unused reclaimed water.

Description of the Production Process and Manufacturing Steps

Alfalfa is considered one of the most important fodder crops grown in desert lands. It is a perennial plant that remains in the land for about 15 years, but its economic feasibility is at the age of 5 years. Clover cultivation is practiced in most desert lands, especially in areas where water is available and cultivated under irrigation system. It is the dominant system of most farming systems

Alfalfa is devided based on several models, including:

1- The models bear the extreme cold and characterized by a period of inactivityin the winter and be completely inactive.
2- The models bear the extreme cold and are characterized by an average period of inactivity during the winter.
3- The models that do not tolerate coldness and are characterized by a lack of dormancy during the winter.

First: Serving the land for agriculture

The central sprinkler is operated on the ground for a limited period, in order to facilitate the operation of the service to the land. Then with the drilled plow the land is plowed twice vertically, taking into account not to plow the places of the tires of the sprinkler, which is used only if there is grass to avoid the milling of Sprayer or cracking in the structure of the central sprinkler after plowing the combs, the ground is left for a long time and then plowing with the discus plow and it is preferred to plow it twice. The drainage is done by circularization, which is not settling but to soften the soil and compress it to facilitate the cultivation process. Dab at the rate of 15 kg per dunum and prefer to hold the disc again after the addition of manure and to stir the fertilizer in the soil. In some cases, the sprinkler is operated by a water cycle at a speed of 70%, in order to increase the moisture content in the soil and therefore not to deepen the cultivation, which is preferred by no more than 3 cm in order to ensure good germination and optimal planting at a depth of 1.5 cm. And also from the benefits of the operation of the sprinkler that the places of agriculture are determined and therefore no loss of part of the seed grown under the tires of the central sprinkler.

Second: alfalfa seeds

There are many sources of alfalfa seeds. There are American varieties such as k101. There are Australian varieties and there are Egyptian varieties. When choosing the seeds, good research is preferred. In order to avoid getting infected with Al-Hamol, a parasitic plant that destroys alfalfa farms, the rate of seedling in the clay lands differs from the desert soil. Seed from 5 to 6 kg per dunum After preparing the land for agriculture, the caliber of the tractor is set to half the quantity of the seed, because it will be done on the farm twice or the cultivation on both sides, even if the cultivation was also half the rate of seedling, it is possible to plant the other half perpendicular on the first agriculture in order to ensure the distribution of seeds on the ground all and ensure the exploitation of the largest amount of land.

Then, after the completion of the planting, the sprayer is operated at 60% speed for three consecutive days and then at the beginning of the bloating of seeds and germination the process is stopped for one day for the arrival of the sun to the seeds and the speed of germination, then the axial sprayer is operated for two days but the speed of operation from 70 to 75% According to the atmosphere of the region, alfalfa is cultivated almost through the whole year in moderate temperate areas, but the best date for planting is the month of March, mid-April and the last half of October to November, and not later than that date to avoid the effects of frost on plants.

The irrigation of alfalfa lasts until the completion of germination and the greenness of the land completely with the crop. and must be careful of the lack of irrigation at this sensitive period when the seedlings emerge from the soil and when the completion of germination in all the land is stopped irrigation for the duration according to the atmosphere and temperature in the region and take care to be sufficient enough to deepen the roots of alfalfa in the soil. The period of thirst is often two days in warm places and can not exceed the week after the period of thirst and then add a quantity of fertillizer and this is the first dose of alfalfa fertillizer and be careful to increase the amount of fertilizer above the required level.

Third: Recommended quantities of fertilizers

5 kg of urea per dunum added in fertilizer plus 2.5 kg of Potassium sulfate compost fertilizer, 2.5 kg of Calcium Sulfate Fertilizer composted and mixed with fertilizer and are well stirred and injected with sprayer. After the addition of this batch of fertilization by about 10 days, the addition of another batch of fertilizers is done with the same rates of fertilizer for the clover (Alfalfa). And it is also added by fertilizer spreader, and this batch is the last batch to the alfalfa before the first frying and is waiting to harvest and the first fry is expected to be the first crop of the weak slightly from the next incense and it can be produce between 75 kg to 120 kg of dried clover (Alfalfa).

Fourth: Harvesting

The Harvesting of the clover (Alfalfa) is done by the harvesters. There are many types of harvesters including self harvesters which are working alone without dragging machine and there aare harvesters that installed in plowing or tractors, also many kinds taken into account when the harvest to be harvested on a suitable height from the surface of ground to keep alfalfa thrones. Preferably at least harvest level of 5 cm the alfalfa is collected in lines behind the harvester and left to drought. Then the alfalfa baler press which clamped by the agricultural tractor doing the alfalfa pressing process carefully and not to run baler press when low humidity to maintain the largest possible number of leaves, where the leaves have the highest protein content, after the alfalfa capsules are sold directly or assembled outside the sprinkler. It works to collect alfalfa and this machine compress the alfalfa on top of each other.

After this period which usually does not exceed the week, a fertilization batch containing 5 urea per dunum, +2.5 kg potassium sulphate per donum, 0.2 kg of other components per dunum, and 1 kg of compound fertilizer which is highly soluble in potassium and phosphorus ratio. The components are mixed well and added to fertilizer. When the fertilizer is added to the fertilizer spreader, the speed of the sprinkler is set to 100%

Technical Requirements Analysis

The main technical requirements for the alfalfa forage crop farming project using central irrigation systems can be summarized as follows:

- A piece of land in the governorate of Ma'an, where the Alfalfa crop is grown.
- Central Pivot irrigation systems with all their equipment and pumps.
- Fixed source of water from underground water wells (or treated water).
- Cutting and pressing equipment according to the latest international standards.
- Power source or generators at the project site.
- Tractors and agricultural equipment.
- Hangars, warehouses, administrative offices and porters.

The Required Land

The project needs a total area of 1000 dunums to grow alfalfa, 500 dunums for each central irrigation system and can allocate 3-4 dunums for services, facilities and aisles, housing for workers, as well as a store for storage of fertilizers and seeds and production requirements and Hanger for agricultural machinery and equipment. It is proposed to rent the land for the project from the princely land in the Ma'an area. The annual rental fee for the dunam is expected to reach no more than 10 JD. The annual land rental cost is estimated at 10,000 JD and is assumed to be an annual increase of 2%.

Construction Works

The project needs the following buildings and facilities: drilling two groundwater wells with depths up to 500 m and a production capacity of not less than (100) cubic meters per hour to reach a production capacity equivalent to 750 thousand cubic meters per year, Housing for workers and hangars and warehouses for materials and mechanisms. The cost includes the construction work required for the project from the design of the building and the plans required for the project and the drilling and paving works as shown in the following table.

Table 12 : The Required Construction Work

Construction Works	Area or quantity (m^2)
Drilling well	2
Well casing	2
Management Offices and Housing	200
Hangers and Warehouses	100
Paving works	400
Others	-

Machinery and Equipment

Includes the cost of the initial investment in addition to the preparing of the site and the work of plowing and leveling of the earth, installation of a system of central irrigation for each 500 dunnums in the form of circular, installation of submersible pumps, underground water pipes, water pumps, tractors 2 complete plowing equipment, Machines for cutting, twisting and drying of the paint in addition to the baler pressing machine and generator of electricity in the event that the area is not covered by the electric grid to turn th central irrigation system on, the following table shows the machines and equipment needed by the project as follows:

Table 13 : The Required Machinery and Equipment

Machinery and Equipment	Quantity
Central Pivot Irrigation System	2
Sub-soil Irrigation network	2
Submersible pumps and accessories	2
Water pumps and pipes	2
Deep, light and discus plows	3
Agricultural equipment and tools	1
Cutting Machine	1
Assembling machine	1
Alfalfa Baler Pressing Machine	1
Large size tractor	1
Medium size tractor	1
Electricity Generator	1
Other Tools	1

Furniture

The project needs simple furniture for management and permanent employment estimated at about 7,000 dinars, which are tables, simple desks, loackers, computers and air conditioners. (Table 14) shows the details of the furniture needed.

Table14 : The Required Furniture

Furniture	Quantity
Offices Furniture	2
Computers	1
Air Conditions	2

Vehicles and Transportation

The project needs a pickup truck to move inside and outside the project as well as to a medium transport vehicle for the project staff and to be used by the marketing representative to reach the target customers.

Table 15 : The Required Transportation and Vehicles

Transportation and Vehicles	Quantity
Pickup Truck	1
Medium transport vehicle	2

Legal Requirements and Procedures

The agricultural activity itself does not require a license, but the fact that the project aims to extract water for the purposes of irrigation and the exploitation of Al-Amiryeh lands for the purposes of agriculture, so must before the start of investment; it must meet a number of legal and administrative conditions. These include legal facility registration, financial registration, obtaining licenses, necessary approvals and others, etc. Delivery of required documents and payment of the initial administrative costs associated with the investment.

1- Legal registration whether it is an individual institution or a company (solidarity, simple recommendation, private contribution, Limited Liability Company, etc.)
2- Registration of the trade name of the company requesting the registration of the trade name.
3- Registration at the Chamber of Industry / Commerce. Every establishment engaged in industrial or commercial activity must be registered in the Chamber of Industry / Commerce.
4- Professional licensing - a prerequisite for legal entities to be able to carry out business activities.
5- Registration at the General Organization for Social Security - Every legal establishment is bound by the provisions of Jordanian law to register for social security.
6- Obtain the approval of the Ministry of Agriculture and Finance to rent land for the implementation of the project
7- The approval of the Ministry of Water and Irrigation shall be obtained for the approval of a pilot for the drilling of groundwater wells

5. Financial Analysis

Assumptions

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.71% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Table 16: Annual Growth Rates Assumptions

	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan , (3) Source: Social Security

Capital Expenditures

The estimated capital costs of the project are 715,646 JD and cover the cost of land, construction, machinery and equipment, furniture and furnishings, vehicles and transportation, pre-operating expenses and initial working capital. These costs are shown in the following tables:

Summary of capital expenditures	Cost (JD)	Contingency	Cost (JD)	Percentage
Construction Works	251,800	10%	276,980	%38.7
Machinery and Equipment	268,100	10%	294,910	%41.2
Furniture	7,000	10%	7,700	%1.1
Vehicles and Transportation	39,000	10%	42,900	%6.0
Pre-operating expenses	22,000	10%	24,200	%3.4
Initial working capital	62,688	10%	68,956	%9.6
Total (JD)	**650,588**		**715,646**	**100%**

Table 17: Construction Work Capital Expenditure

Construction Works	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost / Jordan Dinar
Drilling water well	75,000	2	150,000
Well Works	15,000	2	30,000
Management Offices and Housing	150	200	30,000
Warehouses and Hangers	80	100	8,000
Site Processing and street paving	30	400	12,000
Others	-	-	21,800
Gross Total			**251,800**

Table 18: Machinery and equipment Capital Expenditure

Machinery and equipment	Quantity	Unit cost	Gross Cost /Jordan Dinar
Central Pivot Irrigation System	2	50000	100,000
under soil Irrigation network	2	2500	5,000
Submersible pumps and accessories	2	25000	50,000
Water pumps and pipes	2	5000	10,000
Deep, light and discus plows	3	4000	12,000
Agricultural equipment and tools	1	4000	4,000
Cutting Machine	1	1800	1,800
Assembling machine	1	1200	1,200
Alfalfa Baler Pressing Machine	1	14000	14,000
Large size tractor	1	15000	15,000
Medium size tractor	1	8000	8,000
Electricity Generator	1	25000	25,000
Other Tools	1	-	22,100
Gross Total			**268,100**

Table 19: Furnitutre Capital Expenditure

Furnitutre	Quantity	Unit cost	Gross Cost /Jordan Dinar
Offices furniture	2	2,000	4,000
Computer	1	1,000	1,000
Air Conditions	2	1,000	2,000
Gross Total			**7,000**

Table 20: Transporation and Vehicles Capital Expenditure

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pickup Truck	1	15,000	15,000
Medium transport vehicle	2	12,000	24,000
Gross Total	2		39,000

Table 21: Pre-operating expenses

Pre-operating expenses	Gross Cost
Governmental fees	10,000
Transportation	5,000
Marketing	5,000
Training	2,000
Total	**22,000**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 22: Initial working capital

Initial working capital	Number of months	Percentage	Cost (JD)
Operating expenses	3	%25	46,442
Administrative and general expenses	3	%25	7,741
Marketing expenses	3	%25	505
Indirect salaries	3	%25	8,000
Gross Total			**62,688**

Depreciations

The following tables show all equipment costs, expenses and depreciation rates

:

Table 23: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Construction Works	276,980	%5.0	%0.0
Machinaries	294,910	%10.0	%2.0
Furniture	7,700	%10.0	%5.0
Vehicles	42,900	%10.0	%0.0
	622,490		

Sources of funding

The financing structure required for the project consists of loans and equity. The investment cost of this project was estimated at JD 715,646 JD. This amount is determined through fixed assets, which include land, construction, machinery and equipment, furniture and furnishings, vehicles and transport vehicles, pre-operating expenses and initial working capital. (40%) of the project will be financed through loans (286,259 JD) and the rest (60%) through self financing (equity). The following table shows the general structure of the required financing:

Table 24: Depreciation and annual increases on construction work

Summary of capital costs, additions and annual depreciation	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total fixed assets	622,490	628,773	635,194	641,755	648,459	655,312	662,315	669,474	676,790	684,270
Total consumption expenses	48,400	49,028	49,670	50,326	50,997	51,682	52,383	53,098	53,830	54,578
Total accumulated depreciation	48,400	97,428	147,099	197,425	248,422	300,104	352,487	405,585	459,415	513,993
Total additions	0	6,283	6,420	6,561	6,705	6,852	7,003	7,158	7,317	7,480
Total net book value	574,090	531,345	488,095	444,329	400,037	355,208	309,828	263,888	217,375	170,277

Table 25: Sources and use of finance

Sources of funding	amount	Percentage
Equity (self-financing)	429,388	%60
Loans	286,259	%40
Gross Total	715,646	%100
Use of funding	**amount**	**Percentage**
Capital expenditure	622,490	%86.98
Pre-operating expenses	24,200	%3.38
Working capital	68,956	%9.64
Gross Total	**715,646**	**%100.00**

Table 26: Loan details

	Unit	Value
Loans	Percentage	**40%**
Equity	Percentage	**60%**
Annual Interst Rate	Percentage	**7.50%**
Loans Period	Year	**5**
Loan Starts at year:	Year	**2018**
Grace Period	Year	**1**

Annual Revenues

The selling price per kilogram of the final product (3.75) JD is estimated to increase by 2% annually, as indicated in the market study, based on wholesale prices and consumer prices after deducting profit margin for retailers and wholesalers. The sale of this product to the consumer is between (5.5-6) dinars per kilogram. In general, when estimating the selling price for financial analysis, the price must be very competitive, as the product will compete with existing producers in the local market.

As shown in (Table 25).

Operating expenses

Table (26) shows the details of the costs, starting with agriculture, preparing the land, finishing with the raw materials, packaging and packing supplies for the project.

Administrative and general expenses

Table (27) shows the expected administrative and general expenses of the proposed project (2018-2027) and shows (Table 28) the anticipated marketing expenses of the proposed project. Marketing expenses account for 8% of revenues.

Human Resources

The salaries and monthly wages of each employee were estimated to be multiplied by 16 months of the year for the purpose of calculating the increase in salaries and annual wages due to social security, health insurance, etc. to the employees of the administration. The following table shows the expected annual salaries and wages. The annual rate of increase in wages and salaries is taken into account:

Table 27: Revenues of the project during the years 2018-2027

Descrition	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (tons)	1,250	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500	2,500
Item Price (JD / ton)	280	286	291	297	303	309	315	322	328	335
Total sales (JD)	350,000	714,000	728,280	742,846	757,703	772,857	788,314	804,080	820,162	836,565
Total Project Income (JD)	**350,000**	**714,000**	**728,280**	**742,846**	**757,703**	**772,857**	**788,314**	**804,080**	**820,162**	**836,565**

Table 28: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw material expenses	102,280	204,560	208,651	212,824	217,081	221,422	225,851	230,368	234,975	239,675
Maintenance expenses	0	14,280	14,566	14,857	15,154	15,457	15,766	16,082	16,403	16,731
Consumables	7,000	14,280	14,566	14,857	15,154	15,457	15,766	16,082	16,403	16,731
Direct salaries	59,600	37,904	39,041	40,212	41,419	42,661	43,941	45,259	46,617	48,016
Other expenses	16,888	27,102	27,682	28,275	28,881	29,500	30,132	30,779	31,440	32,115
Gross Total	**185,768**	**298,126**	**304,506**	**311,025**	**317,688**	**324,498**	**331,457**	**338,569**	**345,839**	**353,268**

Table 29: administrative and general expenses

description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Communications expenses	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Hospitality expenses	2,500	2,550	2,601	2,653	2,706	2,760	2,815	2,872	2,929	2,988
Stationery and prints	200	204	208	212	216	221	225	230	234	239
Rent Cost	10,000	10,200	10,404	10,612	10,824	11,041	11,262	11,487	11,717	11,951
Consulting Costs	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Training Cost	1,000	0	0							
Other expenses	12,450	12,699	12,953	13,212	13,476	13,746	14,020	14,301	14,587	14,879
Gross Total	**2,815**	**2,769**	**2,825**	**2,881**	**2,939**	**2,998**	**3,058**	**3,119**	**3,181**	**3,245**

Table 30: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	2,018	2,058	2,100	2,142	2,184	2,228	2,273	2,318	2,364	2,412

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project Manager -Agricultural Engineer/Food Processing	12,800	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Agricultural Irrigation Technician	8,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Marketing representative and accountant	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Guard	4,800	4,944	5,092	5,245	5,402	5,565	5,731	5,903	6,080	6,263
Total indirect salaries	**32,000**	**32,960**	**33,949**	**34,967**	**36,016**	**37,097**	**38,210**	**39,356**	**40,537**	**41,753**
Direct Employees										
Maintenance Technician	6,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Cultivation workers	9,600	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Tractors Driver	12,000	16,480	16,974	17,484	18,008	18,548	19,105	19,678	20,268	20,876
Total direct salaries	**27,600**	**37,904**	**39,041**	**40,212**	**41,419**	**42,661**	**43,941**	**45,259**	**46,617**	**48,016**
Gross Total	**59,600**	**70,864**	**72,990**	**75,180**	**77,435**	**79,758**	**82,151**	**84,615**	**87,154**	**89,768**

Income Statement

Table 31: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	350,000	714,000	728,280	742,846	757,703	772,857	788,314	804,080	820,162	836,565
Gross Operating Revenues	350,000	714,000	728,280	742,846	757,703	772,857	788,314	804,080	820,162	836,565
Operating Expenses	(185,768)	(298,126)	(304,506)	(311,025)	(317,688)	(324,498)	(331,457)	(338,569)	(345,839)	(353,268)
Gross Operating Profit	164,232	415,874	423,774	431,820	440,014	448,359	456,857	465,511	474,323	483,297
Gross Profit Percentage	%47	%58	%58	%58	%58	%58	%58	%58	%58	%58
Salaries and Benefits (Indirect Staff)	(32,000)	(32,960)	(33,949)	(34,967)	(36,016)	(37,097)	(38,210)	(39,356)	(40,537)	(41,753)
General and Administraive Expenses	(30,965)	(30,462)	(31,071)	(31,693)	(32,327)	(32,973)	(33,633)	(34,305)	(34,991)	(35,691)
Markeing Expenses	(2,018)	(2,058)	(2,100)	(2,142)	(2,184)	(2,228)	(2,273)	(2,318)	(2,364)	(2,412)
Pre Operating Expenses	(24,200)									
Gross Indirect Expenses	(89,183)	(65,480)	(67,120)	(68,802)	(70,527)	(72,298)	(74,115)	(75,979)	(77,892)	(79,856)
Income before Interest, Depreciation and Tax	75,049	350,393	356,654	363,019	369,487	376,061	382,742	389,531	396,431	403,441
Fixed Assets Depreciations	(48,400)	(49,028)	(49,670)	(50,326)	(50,997)	(51,682)	(52,383)	(53,098)	(53,830)	(54,578)
Income before Tax and Interests	26,649	301,365	306,984	312,692	318,490	324,379	330,359	336,433	342,601	348,863
Bank Interests	(21,469)	(17,773)	(13,800)	(9,528)	(4,936)	0	0	0	0	0
Income Before Tax	5,180	283,592	293,184	303,164	313,554	324,379	330,359	336,433	342,601	348,863
Income Tax	(259)	(14,180)	(14,659)	(15,158)	(15,678)	(16,219)	(16,518)	(16,822)	(17,130)	(17,443)
Net Profit	4,921	269,412	278,525	288,006	297,876	308,160	313,841	319,611	325,470	331,420

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit Percentage	%1	%38	%38	%39	%39	%40	%40	%40	%40	%40
Compulsory Reserves	(492)	(26,941)	(27,853)	(28,801)	(29,788)	(30,816)	(31,384)	(31,961)	(32,547)	(33,142)
Retained Earnings	4,429	246,900	497,572	756,778	1,024,866	1,302,210	1,584,667	1,872,318	2,165,241	2,463,519

Expected Balance Sheet

Table 32: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	30,337	220,324	482,425	750,196	1,023,717	1,373,819	1,730,094	2,092,642	2,461,561	2,836,954
Inventory	8,523	17,047	17,388	17,735	18,090	18,452	18,821	19,197	19,581	19,973
accounts receivable	58,333	119,000	121,380	123,808	126,284	128,809	131,386	134,013	136,694	139,427
Total assets traded	**97,194**	**356,371**	**621,193**	**891,739**	**1,168,091**	**1,521,080**	**1,880,301**	**2,245,852**	**2,617,836**	**2,996,354**
Non-current assets										
Fixed assets (net)	574,090	531,345	488,095	444,329	400,037	355,208	309,828	263,888	217,375	170,277
Total non-current assets	**574,090**	**531,345**	**488,095**	**444,329**	**400,037**	**355,208**	**309,828**	**263,888**	**217,375**	**170,277**
Total assets	**671,284**	**887,716**	**1,109,288**	**1,336,069**	**1,568,128**	**1,876,288**	**2,190,129**	**2,509,741**	**2,835,211**	**3,166,631**
Liabilities	-	-	-	-	-					
Current liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Payables	0	0	0	0	0	0	0	0	0	0
Remaining amount of Loan	52,980	56,953	61,225	65,817	0	0	0	0	0	0
Total current liabilities	**52,980**	**56,953**	**61,225**	**65,817**	**0**	**0**	**0**	**0**	**0**	**0**
Non-current liabilities										
long term loans	183,995	127,042	65,817	0	0	0	0	0	0	0
Total long - term liabilities	**183,995**	**127,042**	**65,817**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total liabilities	**236,975**	**183,995**	**127,042**	**65,817**	**0**	**0**	**0**	**0**	**0**	**0**
Property rights										
Members Contributions	429,388	429,388	429,388	429,388	429,388	429,388	429,388	429,388	429,388	429,388
Compulsory reserve	492	27,433	55,286	84,086	113,874	144,690	176,074	208,035	240,582	273,724
Profit (loss)	4,429	246,900	497,572	756,778	1,024,866	1,302,210	1,584,667	1,872,318	2,165,241	2,463,519
Total equity	434,309	703,721	982,246	1,270,252	1,568,128	1,876,288	2,190,129	2,509,741	2,835,211	3,166,631
Total liabilities and equity	**671,284**	**887,716**	**1,109,288**	**1,336,069**	**1,568,128**	**1,876,288**	**2,190,129**	**2,509,741**	**2,835,211**	**3,166,631**
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Expected Cashflows Statement

Table 33: Expected Cashflows Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	4,921	269,412	278,525	288,006	297,876	308,160	313,841	319,611	325,470	331,420
Non-monetary adjustments										
Bank Interests	21,469	17,773	13,800	9,528	4,936	0	0	0	0	0
Depreciation	48,400	49,028	49,670	50,326	50,997	51,682	52,383	53,098	53,830	54,578
Total Operating Cashflows before	74,790	336,214	341,995	347,860	353,809	359,842	366,224	372,710	379,301	385,998
Additions to Working Capital										
Inventory (Increase/Decrease)	(8,523)	(8,523)	(341)	(348)	(355)	(362)	(369)	(376)	(384)	(392)
Accounts Receivables (Increase/Decrease)	58,333)	(60,667)	(2,380)	(2,428)	(2,476)	(2,526)	(2,576)	(2,628)	(2,680)	(2,734)
Accounts Paybales (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Working Capital (Increase/Decrease)	(66,857)	(69,190)	(2,721)	(2,775)	(2,831)	(2,887)	(2,945)	(3,004)	(3,064)	(3,125)
Net Cashflows from Operating Activities	7,934	267,024	339,274	345,085	350,978	356,955	363,279	369,706	376,236	382,872
Cashflows from Investments Activities										
Fixed Assets (Procurement)	622,490)	(6,283)	(6,420)	(6,561)	(6,705)	(6,852)	(7,003)	(7,158)	(7,317)	(7,480)
Net Cashflows from Investments Activities	622,490)	(6,283)	(6,420)	(6,561)	(6,705)	(6,852)	(7,003)	(7,158)	(7,317)	(7,480)
Cashflows from Financing Activities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Capital	429,388									
Loan Amortization	(49,284)	(52,980)	(56,953)	(61,225)	(65,817)	0	0	0	0	0
Bank Interest Rate	(21,469)	(17,773)	(13,800)	(9,528)	(4,936)	0	0	0	0	0
Loans	286,259	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	644,893	(70,753)	(70,753)	(70,753)	(70,753)	0	0	0	0	0
Net increase (decrease) in cash	30,337	189,987	262,101	267,771	273,521	350,102	356,275	362,547	368,919	375,393
Cashflows at the Beginning of Period	0	30,337	220,324	482,425	750,196	1,023,717	1,373,819	1,730,094	2,092,642	2,461,561
Cashflows at the end of Period	30,337	220,324	482,425	750,196	1,023,717	1,373,819	1,730,094	2,092,642	2,461,561	2,836,954

Feasibility Indicators

In order to access the discounted cash flow statement, the discount factor must be calculated where there is more than one method used to calculate it, but in the study period the weighted average cost of capital (WACC) is used as a discount factor to calculate the discounted cash flows and net present value of the investment. The financing structure of the project is based on equity and loans. Therefore, the discount rate should be adjusted according to the weighted average cost of capital (WACC). This is based on the assumptions shown in Table (32):

Table 34: Risk Premium

	Rate
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premuim	9.93%

Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 35: Weighted Average Cost of Capital

Weighted Average Cost of Capital	Indicator
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premuim	%9.93
Income Tax Rate	%5
Bets	1.00
Equity	429,388
Loans Value	286,259
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.13
Loans Value	286,259
Loans Percentage	%40
Equity	

Weighted Average Cost of Capital	Indicator
Cost of Equity	%16.4
Equity Value	429,388
Equety Percentage	%60
Gross	
Project Value	715,646
Weighted Average Cost of Capital	%12.71

The following table illustrates the net expected free cash flow of the project:

Table 36: Free Net Cashflow

Net Cashflows	Pre-Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	(715,646)	32,134	260,740	332,854	338,524	344,274	350,102	356,275	362,547	368,919	375,393	4,483,332
Discount Factor	1.00	0.89	0.79	0.70	0.62	0.55	0.49	0.43	0.38	0.34	0.30	0.30
Net Present Value for Cash Flows	**(715,646)**	**28,510**	**205,257**	**232,482**	**209,783**	**189,291**	**170,792**	**154,206**	**139,228**	**125,701**	**113,485**	**1,355,356**

Table 37 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	Rate	%12.71
Net Present Value for Cashflows	JD	2,208,446
Payback Period	Year	3
Internal Rate of Return	Rate	%38.50